Replicating Manifold Steering: Geometry of LLM Representations and Behavior

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Code, checkpoints, and figures: https://github.com/kirillacharya/manifold-steering-replication

Why geometry?

Large language models build up internal representations layer by layer: at layer L, each token maps to a hidden state $h^{(L)} \in \mathbb{R}^d$, and structured concept domains may organize into geometric patterns. Hidden states for *days of the week* or *months of the year* may trace a smooth, low-dimensional manifold $\mathcal{M}_h \subset \mathbb{R}^d$ — closed loops for cyclic domains, open curves for sequential ones. Wurgaft et al. [1] ask whether this geometry *predicts behavior*: output probability vectors, embedded via the Hellinger map $p \mapsto \sqrt{p}$ (which makes the Hellinger distance $d_H(p,q) = \frac{1}{\sqrt{2}}\|\sqrt{p} - \sqrt{q}\|$ Euclidean [2]), may trace a matching *behavior manifold* $\mathcal{M}_y$. If $\mathcal{M}_h \approx \mathcal{M}_y$ isometrically, steering *along* $\mathcal{M}_h$ should produce natural behavioral transitions, while linear steering cuts off-manifold and "teleports" probability mass. The two strategies interpolate between a source concept $\hat{h}_{c_s}$ and a target $\hat{h}_{c_e}$:

$$\pi_{\mathrm{lin}}(\alpha) = (1-\alpha)\,\hat{h}_{c_s} + \alpha\,\hat{h}_{c_e}, \qquad \pi_{\mathrm{mfd}}(\alpha) = s\big((1-\alpha)\,u_{c_s} + \alpha\,u_{c_e}\big), \quad u_{c_i} = s^{-1}(\hat{h}_{c_i}),$$

where $s : \mathbb{R}^k \to \mathbb{R}^d$ maps intrinsic manifold coordinates back to activation space. This report replicates the paper's central figure on a small open model, then extends the idea to a genuinely two-dimensional concept structure (a day×hour torus).

The experiment, step by step

Setup. All experiments use instruction-tuned **Gemma-2-2B** ($d = 2304$), patching layer 24 of 26. Two cyclic concept domains: *weekdays* (7 classes) and *months* (12 classes), probed with arithmetic prompts — "`Q: What day is two days after Monday? A:`" (answer: *Wednesday*), 6 prompts per concept (`scripts/run_weekdays.py`, `run_months.py`). For each prompt we record the last-token hidden state $h^{(24)}$ and the output distribution over concept tokens, then average per concept: $\hat{h}_c = \frac{1}{N_c}\sum_{i\in S_c} h_i^{(L)}$ and $\hat{p}_c = \frac{1}{N_c}\sum_{i\in S_c} p_i$.

Manifolds. Activation centroids $\{\hat{h}_{c_k}\}$ are reduced by PCA to r dimensions (r=6 weekdays, r=11 months) and a cubic spline γ with C^2 continuity is fitted through them in concept order, giving $\mathcal{M}_h$; the same procedure on Hellinger-embedded behavior centroids $\{\sqrt{\hat{p}_{c_k}}\}$ gives $\mathcal{M}_y$.

Steering. For $\alpha \in [0,1]$ between spline indices i_s, i_e: $\pi_{\mathrm{mfd}}(\alpha) = \mathrm{PCA}^{-1}\big(\gamma(i_s + \alpha\,\Delta i)\big)$. At each of 30 steps, a forward hook replaces the layer-24 hidden state at the last token with $\pi(\alpha_t)$ and the output probability of every concept token is recorded (`scripts/run_steering_probs.py`). The pipeline is deterministic (no sampling, no training); every figure below regenerates from committed checkpoints without a GPU, and collection scripts expose `-seed` for exact reproducibility.

Replication of the main figure

Figure 1 replicates the paper's main result. In both domains the concept centroids trace closed, ordered loops in activation space and in behavior space — the cyclic structure of weeks and months is visibly encoded. The manifold path stays on the fitted curve and, in the probability trajectories (row 3), produces *smooth, ordered* transitions: steering from Tuesday to Friday passes probability mass through Wednesday and Thursday in sequence. Linear steering instead leaves the manifold and *teleports*: intermediate concepts never rise in order, and off-concept mass appears mid-path. One practical lesson: the patch must be applied deep enough. At layer 12, keys and values computed from the original prompt override the patched query over the 14 remaining blocks and both methods flatline; at layer 24 (92% depth) the replication is clean.

Extension: torus steering (day × hour)

Do LLMs encode *two-dimensional* concept structure? We collect layer-24 activations for all 168 prompts "`It is HH:00 on {Day}`" (7 days × 24 hours) and recover two 2D subspaces $V_{\mathrm{day}}, V_{\mathrm{hour}} \in \mathbb{R}^{2\times d}$ by ordinary least squares onto $[\sin\theta, \cos\theta]$ labels, QR-orthogonalized (`scripts/run_hierarchical_steering.py`). Projecting every activation onto both subspaces and embedding on a 3D torus surface (Fig. 2, left) reveals 7 ordered day-arcs; the hour coordinate is only partially circular at this layer.

Steering in *product coordinates* moves each factor independently: $h(\alpha) = h_{\mathrm{base}} + \Delta_{\mathrm{day}}(\alpha)\,V_{\mathrm{day}} + \Delta_{\mathrm{hour}}(\alpha)\,V_{\mathrm{hour}}$. Steering from $(\mathrm{Tue}, 09{:}00)$ to $(\mathrm{Fri}, 15{:}00)$ over six steps (Fig. 2, right), torus steering shifts probability mass smoothly in *both* dimensions, while linear steering moves the day correctly but teleports the hour, jumping to 15:00 only at the final

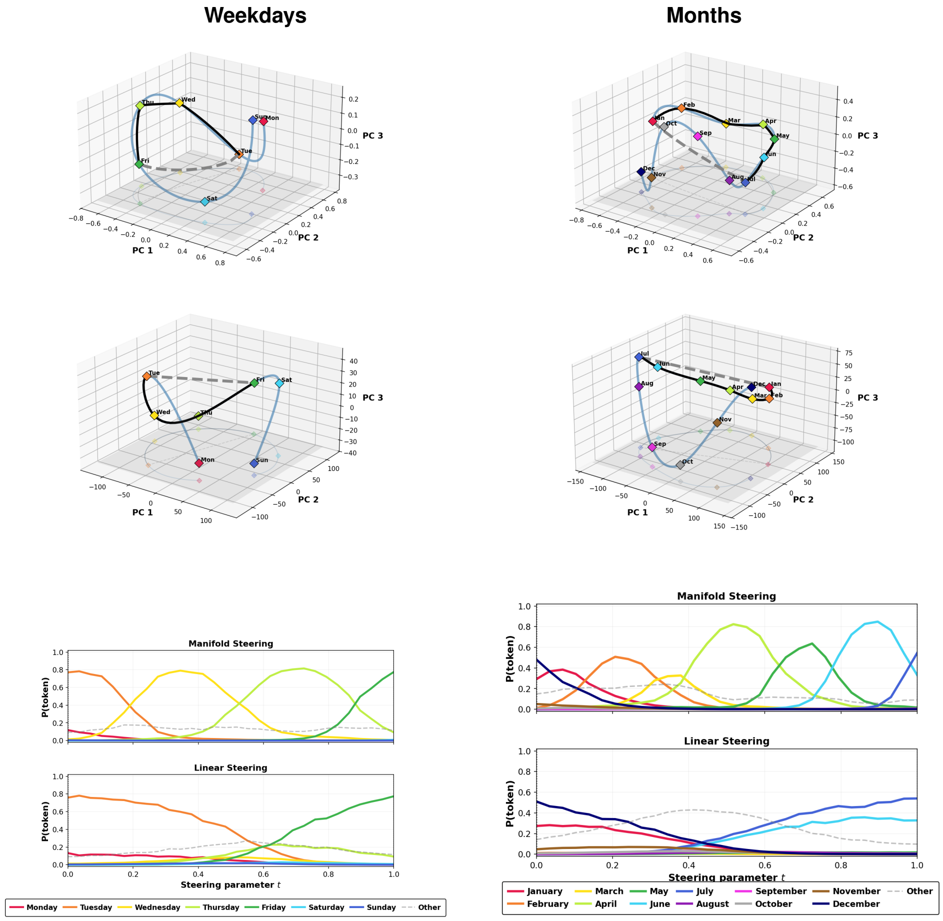

Figure 1: Gemma-2-2B layer 24, steering Tue→Fri (weekdays) and Jan→Jul (months). Rows 1–2: fitted behavior manifolds $\mathcal{M}_y$ (Hellinger–PCA) and activation manifolds $\mathcal{M}_h$ (activation–PCA) with the linear path (dashed) and manifold path (black). Row 3: concept-token probabilities along each path.

step — the same off-manifold failure mode as in the 1D domains, now visible per coordinate.

Takeaway. Both the replication and the torus extension support the paper's claim: the geometry of a model's representations is not decoration — moving along it is what produces natural behavior, and ignoring it (linear steering) breaks transitions in exactly the way the manifold picture predicts. All code, checkpoints, SLURM scripts, and a no-GPU figure regeneration path are at https://github.com/kirillacharya/manifold-steering-replication.

References

[1] D. Wurgaft, C. Rager, M. Kowal, V. Shyam, S. Feucht, U. Bhalla, T. Haklay, E. Bigelow, R. Sarfati, T. McGrath, O. Lewis, J. Merullo, N. D. Goodman, T. Fel, A. Geiger, and E. S. Lubana. Manifold steering reveals the shared geometry of neural network representation and behavior. *arXiv preprint arXiv:2605.05115*, 2026.

[2] S. Amari and H. Nagaoka. *Methods of Information Geometry.* American Mathematical Society, 2000.

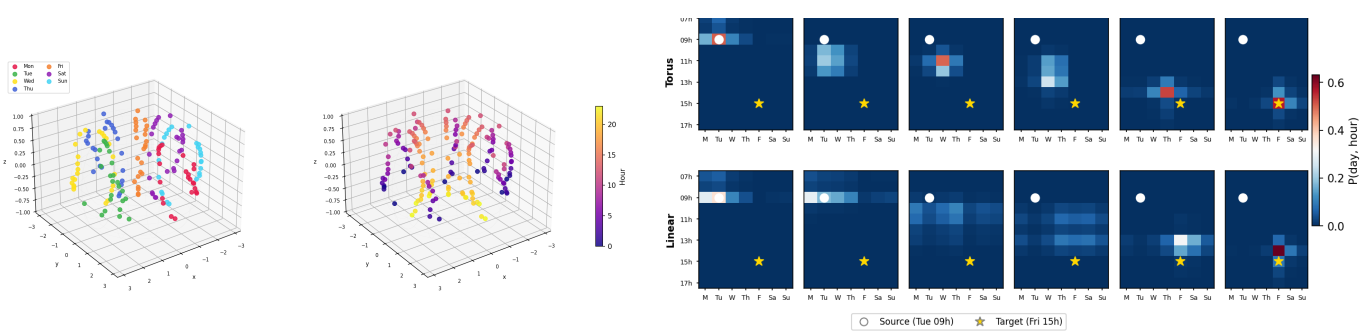

Figure 2: Left: layer-24 activations on the day×hour torus, colored by day (7 ordered arcs) and by hour. **Right:** day×hour probability grids across 6 steering steps, (Tue, 09:00) → (Fri, 15:00); top row torus (product-coordinate) steering, bottom row linear steering.